INCH-POUND
MIL-DTL-32155 (OS)
06 October 2004

MILITARY SPECIFICATION

THRUSTER, CARTRIDGE ACTUATED TCU-5A/A

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the TCU-5A/A Thruster, Cartridge Actuated (Thruster) used for seat drop operation on the V-22 Emergency Escape Hatch Removal System (EEHRS).

1.2 Application. This Thruster was developed for use in the Emergency Egress Hatch Removal System as the seat drop Thruster and is used at the three cabin hatch locations on V-22 Aircraft, and its variants. It drops the seats to ensure a clear egress path when the EEHRS is activated. No changes to the configuration will be permitted without the authorization from Indian Head Division, Naval Surface Warfare Center, Code 50.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Branch (Code 4230), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, or emailed to pennam@ih.navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-D-23615	Design and Evaluation of Cartridge Actuated Devices

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-1168	Ammunition Lot Numbering and Ammunition Data Cards

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

3205AS117-2	Thruster, Cartridge Actuated TCU-5A/A
3205AS111	Cord, Detonating Assembly

(Application for copies should be addressed to the Commanding Officer, Naval Air Technical Service Facility, 700 Robbins Avenue, Philadelphia, PA 19111-5079.)

2.3 Non-Government publications and drawings. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

BELL-BOEING DOCUMENTS (CAGE Code 77272)

901-947-784	Emergency Escape Hatch Removal System Specification
901-947-034	V-22 Vibration Test Report

(Application for copies of Bell-Boeing documents should be sent to Boeing Defense and Space Group, Helicopter Division, P.O. Box 16858, Philadelphia, PA 19142-0858)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

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ASTM E 1742 Standard Practice for Radiographic Inspection

(Copies of this document are available online at http://www.astm.org/cgi-bin/SoftCart.exe/STORE/standardsearch.shtml?L+mystore+afpt6301+1062008944 or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

Universal Propulsion Company (CAGE Code 17610)

17682-1 Functional test firing setup

(Application for copies of this document should be addressed to the Universal Propulsion Company, Goodrich Corporaton, 3530 Branscombe Road, P.O. Box KK, Fiarfield, CA 94533-0659.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to the first article inspection, Table I, and the first article test plan in Table II, as specified in sections 4.2.1 and 6.3. A first article inspection shall be required if two years or more has passed since the Thruster or a similar item were manufactured, major turn over in personnel, major changes to the production facility or transfer of production to a new facility or site.

3.2 Flight critical part. The Thruster's are flight critical items, as defined in 6.4.2, and relate to personnel safety and emergency egress and as such the technical data package shall be identified and procured as such.

3.3 Design and construction. The following applies to the qualification status and configuration control of the Thrusters over the entire life of the item.

3.3.1 Identical configuration. The Thruster's design and construction shall be identical to that subjected to qualification testing as documented on 3205AS117, and approved for service use by the Navy.

3.3.2 Configuration control. This item is a configuration-controlled device. Only the Thruster's procured from the vendor(s) listed on drawing 3205AS117 are approved for the application specified

3.4 Primary components. Primary components are components specifically manufactured for this Thruster and not commercially available off-the-shelf items. Only one lot of each primary component shall be used in a Thruster lot. One primary lot may be used in more than one Thruster lot.

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3.5 Thruster production. The Thrusters shall be manufactured in accordance with drawings listed on 3205AS117. Each production Thruster shall meet the requirements of 3.4 and 3.5.1 through 3.5.2.4 as outlined in Table I. Failure of any Thruster to meet the requirements of 3.4 and Table I shall result in rejection of the lot of Thrusters.

3.5.1 Visual inspection. The Thruster shall be free of the following defects: illegible, missing or inaccurate markings, damage, burrs, dents, sharp edges, chipped or incomplete protective finish, or other defects which may prevent the installation of the Thruster. Each Thruster shall meet the dimensional and finish requirements of the drawings 3205AS117.

3.5.2 In-process inspections. The following are the minimum in-process inspections required for product integrity.

3.5.2.1 Corrosion. All components scheduled for use in production shall be free of corrosive products (see 4.1.4)

3.5.2.2 X-ray examination. Each Thruster shall be x-rayed along the longitudinal axis. When radiographs are examined in accordance with 4.3.2, the Thruster shall show proper assembly; presence of all parts, and sealing in accordance with approved radiographic inspection criteria. The radiographic inspection shall show no foreign materials present. All defects shall be identified on the film and defective Thrusters removed from the lot.

3.5.2.3 N-ray examination. Each Thruster shall be N-rayed along the longitudinal axis. When radiographs are examined in accordance with 4.3.2, the Thruster shall show proper order, presence and condition of parts in accordance with approved radiographic inspection criteria. All defects shall be identified on the film and defective Thruster's removed from the lot.

Table I. First Article /Production inspections.

Test Sequence	Test Paragraph	Requirement Paragraph
1. Visual	4.3.1	3.5.1
2. Radiographic X-ray Inspection	4.3.2	3.5.2.2
3. Radiographic N-ray Inspection	4.3.2	3.5.2.3

3.6 Thruster First Article Test environmental exposure. The Thruster shall meet all requirements of 3.6.1 through 3.6.5 and the acceptance criteria of 4.4.

3.6.1 Vibration. The Thruster shall not function or incur any internal/external damage or degradation when vibration cycles are applied in accordance with 4.3.3. The Thruster shall be x-ray inspected per 3.5.2.2 after completion of vibration. Thrusters will then be subjected to shock testing per 3.6.2.

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3.6.2 Shock. The Thruster shall not function or incur any internal/external damage or degradation when saw tooth shocks are applied in accordance with 4.3.4.Thrusters will then be subjected to x-ray inspection per 3.5.2.2 and then functionally tested per 3.6.5.

3.6.3 Temperature-shock/humidity/altitude (TSHA) cycling. The Thruster shall not function or incur any internal/external damage or degradation when subjected to TSHA cycling in accordance with 4.3.5. The Thrusters will be visually inspected per 3.5.1 and then subjected to functional testing per 3.6.5.

3.6.4 Six-foot drop. The Thruster shall not stroke or incur any major internal/external damage or degradation as a result of the six-foot drop test in accordance with 4.3.6. The Thrusters will be visually inspected per 3.5.1 and then subjected to functional testing per 3.6.5.

3.6.5 Functioning. The Thruster shall meet the requirements of 3.6.5.1 through 3.6.5.3, when fired in accordance with 4.3.7.

3.6.5.1 Stroke. The Thruster shall break the shear pin and overcome the 197 lbs. (-0/+1) pre-load, producing a stroke of 0.45-inches minimum across the temperature range per 4.3.7.1.

3.6.5.2 Peak thrust. The peak thrust shall not exceed 600 lbs. for duration longer than 5 milliseconds across the operational temperature range per 4.3.7.2.

3.6.5.3 Time to stroke. The time to full stroke (0.45-inch) shall not exceed 1.0 second across the operational temperature range.

3.7 Workmanship. The Thruster shall be constructed and finished in a manner to assure compliance with all requirements of this specification and associated drawings.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1)
b. Production inspection (see 4.2.2)
c. Lot acceptance inspection (see 4.2.3)

4.1.1 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.2.1, 4.2.2 and 4.2.3.

4.1.2 Interchangeability. The contractor shall conduct inspections of the production lot to verify total interchangeability of the mounting provisions and port connection.

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4.1.3 Primary components. Primary components as identified in 3.4, shall be verified by review of Data Cards submitted as contract deliverables.

4.1.4 Corrosion. To meet the requirements of paragraph 3.5.2.1, all parts shall be visually inspected to assure there are no corrosion products on the hardware

4.2 Inspections.

4.2.1 First article inspection. Unless otherwise specified in the contract or purchase order (see 6.2), a first article inspection sample of 21 Thrusters conforming to drawing 3205AS117, shall be subjected to first article testing. These Thrusters shall be inspected to the requirements of Table I. These Thrusters shall be expended in the tests listed in Table II. Any damage inflicted by the environmental exposures which would adversely affect the performance or service life of the item in service application shall be cause for rejection of the first article sample. Any production prior to notification by the contracting agency of the first article sample acceptability shall be at the contractor's risk. Failure of any Thruster to comply with the requirements of section 3 shall be cause for rejection of the first article represented.

4.2.2 Production inspection. To meet the requirements of 3.5, all production Thruster manufactured under the contract shall be inspected and screened for defects. Thruster's failing to meet the requirements listed in Table I shall be rejected and removed from the lot.

4.2.3 Lot acceptance inspection. Lot acceptance inspection shall consist of the examinations and tests specified in Table III. Failure of any sample Thruster to comply with the requirements listed in Table III shall be cause for rejection of the lot represented. Packaging defects shall be corrected before acceptance.

TABLE II. First article test plan.

Test Sequence	Test Para.	Req. Para.	Sample Groups						
			I	II	III	IV	V	VI	VII
1. Visual	4.3.1	3.5.1	6	3	3	3	3	3	3
2. Radiographic X-ray	4.3.2	3.5.2	6	3	3	3	3	3	3
3. Radiographic N-ray	4.3.2	3.5.2	6	3	3	3	3	3	3
4. Vibration	4.3.3	3.6.1	6						
5. Shock	4.3.4	3.6.2	6						
6. TSHA	4.3.5	3.6.3		3					
7. 6 Foot Drop	4.3.7	3.6.4			3				
8. Spares									3
9. Function test:	4.3.7	3.6.5							
-65°F ± 5°F			2	1	1	3			
70 ± 5°F			2	1	1		3		
200 ± 5°F			2	1	1			3	

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Table III. Lot acceptance inspection and tests.

Inspection/Test	Test Paragraph	Requirement Paragraph	Quantity
1. Visual Inspection	4.3.1	3.5.1	100%
2. Radiographic X-ray Inspection	4.3.2	3.5.2.3	100%
3. Radiographic N-ray Inspection	4.3.2	3.5.2.2	100%
4. Functional test: -65 ± 5°F 70 ± 5°F 200 ± 5°F	4.3.7	3.6.5	See Table II or Table IV

4.2.3.1 Sample size. A random sample of Thruster's from each production lot shall be selected in accordance with Table IV for lot acceptance inspection. Test samples shall not be applied as part of the quantity specified for delivery by the contract or purchase order. Lot acceptance for variable data is based on all parameters meeting the requirements and the process capability ratio, Cpk, being greater than 1.0 at each conditioning temperature.

Table IV. Lot acceptance sampling.

Lot Size	Number of Test Samples		
	-65F	+70F	+160F
5-50	3	3	3
51-90	4	4	4
91-150	5	5	5
151-200	6	6	6
201-500	7	7	7

4.3 Inspections and tests.

4.3.1 Visual inspections.

4.3.1.1 Thruster inspection. The external condition and appearance of the Thruster shall be determined by comparison to drawings NAVAIR 3205AS117. Specific inspections of the warning label, TLX port, mounting provisions, proper color, finish and correct data on label shall be conducted at a minimum. Each Thruster shall meet the requirements of 3.5.1 and 3.7.

4.3.1.2 Packaging inspection. The condition of the packaging (inner container), packing (outer container), and markings shall be determined by comparison to the requirements of section 5.

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4.3.2 Radiographic examination. Radiographic examination shall be conducted in accordance with ASTM-E 1742. All Thrusters shall be identified with serial numbers beginning with 001 prior to examination. The Thruster shall be arranged on boards or trays in consecutive order, with any missing serial numbers identified on the radiographic plate. Each radiograph shall carry a permanent identification of the items displayed thereon in a 4 x 6-inch region maximum. The radiographic identification shall include the drawing number, the complete lot number in accordance with MIL-STD-1168, the contract number, and the span of serial numbers displayed. Radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the lot acceptance tests. Any observable imperfections as outlined in 3.5.2.2 and 3.5.2.3, shall be cause for rejection of the Thruster. Defective Thrusters found during radiographic review are to be marked on the radiographic plate and removed from the production lot. Setting on the radiographic equipment shall be noted on the radiographic review sheets.

4.3.3 Vibration. Per 901-947-784 and 901-947-034 the Thruster shall be subjected to the following vibration profiles: 2 units at -65°F, 2 units at 70°F, and 2 units at 160°F. Each subscale Thruster shall be subjected to the combined source dwell and random excitation environments or individual source dwell and random excitation environments as applicable at -65°F, +70°F and +160°F in each axis for 20 minutes.

a. Conduct a resonance survey for each Thruster configuration in each of the three mutually perpendicular axes both prior to and following exposure to the source dwell environment. Conduct a low level (0.5 g peak), 5-500 Hz sweep at a frequency rate of 1/2 octave per minute (15 minutes duration) with a tolerance equal to $\pm$ 10% on the vibration amplitude and $\pm$ 2% or 1 Hz, to determine the Thruster configuration's resonant frequency and transmissibility. The same Thruster of each configuration and same mounting location on the vibration fixture will be utilized for both the pre- and post-source dwell frequency surveys. The resonant frequency shall be considered to be any natural frequency with an amplification factor of 2.0 or greater output response relative to the input level. The Thruster response shall be measured on the least supported location or a location shown to have significant amplification. All resonance surveys will be conducted at ambient temperature. Following the completion of the source dwell vibration testing at -65°F and 160°F, the Thruster must be returned to ambient temperature prior to completion of the post-source dwell resonance survey.

b. Predicated upon the outcome of the resonance survey, source dwell vibration and random vibration excitation are to be conducted simultaneously. If based upon the resonance survey, the four Government desired test frequencies exceed the permitted test frequencies and the source dwell vibration and random excitation cannot be applied simultaneously, the source dwell vibration levels shall be increased by 0.5 g.

Source dwell vibration shall be performed at each of the four amplitudes and frequencies established by the Government based upon the resonance survey along each of the three- (3) mutually perpendicular axes. Source dwell input levels are shown in figure 1 and shall include the superimposed broad band random background levels of the random excitation as described in 4.3.4.c.

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c. Random excitation shall be conducted in accordance with MIL-STD-810, Method 514.3, Paragraph I-4.2.2, and Technique 3. Random excitation shall be a flat spectrum at 0.02 g^2/Hz over the bandwidth from 10-2000 Hz as shown in figure 2.

d. The following tolerances apply to all vibration testing:

(1) Instantaneous random vibration acceleration peaks shall be limited to three (3) times the RMS acceleration (3-sigma clipping).

(2) The random vibration spectrum shall be within the tolerance range of $\pm$ 3 dB between 10 and 2000 Hz except deviations of $\pm$ 6 dB are permissible over a cumulative bandwidth of 200 Hz maximum between 500 and 2000 Hz.

(3) Overall RMS acceleration must be within $\pm$ 1.5 dB of the nominal input level.

(4) Sine tones must be within $\pm$ 10% of the nominal value.

4.3.4 Shock. Per 901-947-784 the Thruster with protective caps installed in the manifold ports, shall be submitted to the shock test outlined in MIL-STD-810, Method 516.3 Procedure I to the levels in Figure 4, three (3) shocks in each direction along each of the three (3) axis for a total of eighteen (18) shocks. All shocks will be conducted at ambient temperature.

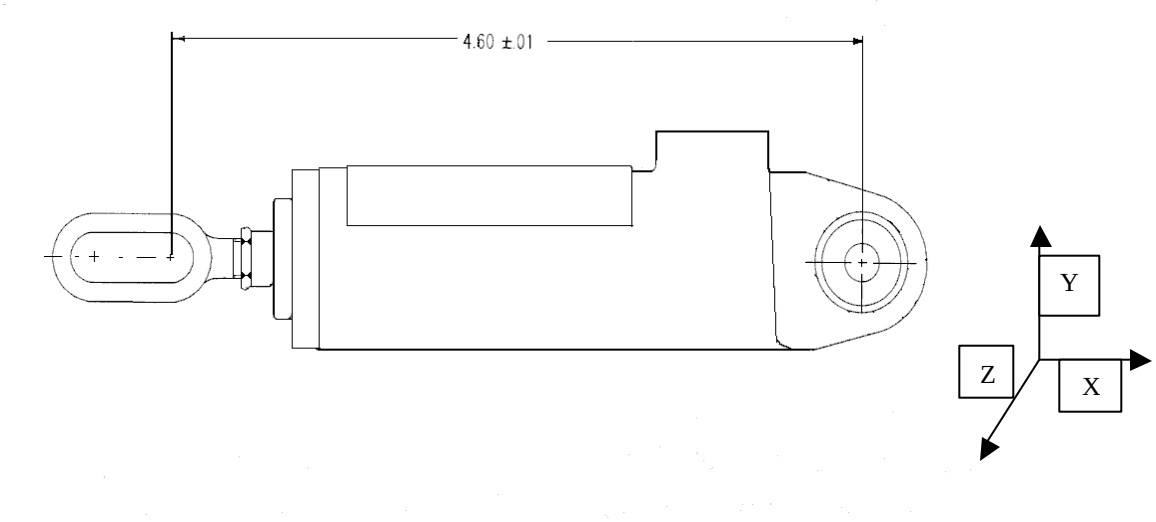


Figure 1. Shock / Vibration and Drop Axis

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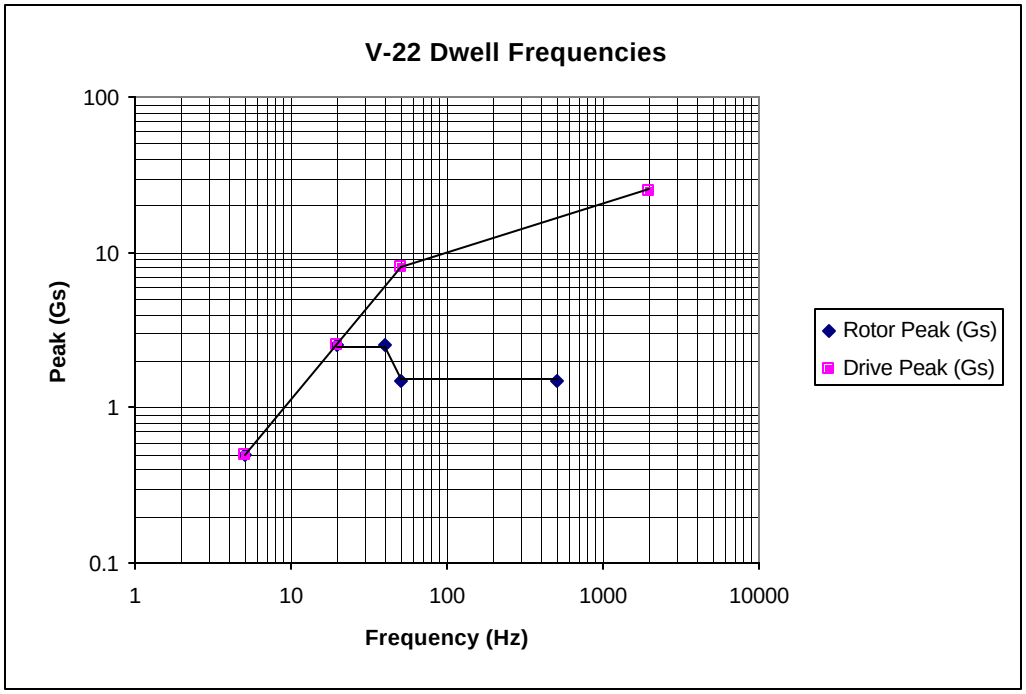


Figure 2. Source Dwell Input Levels

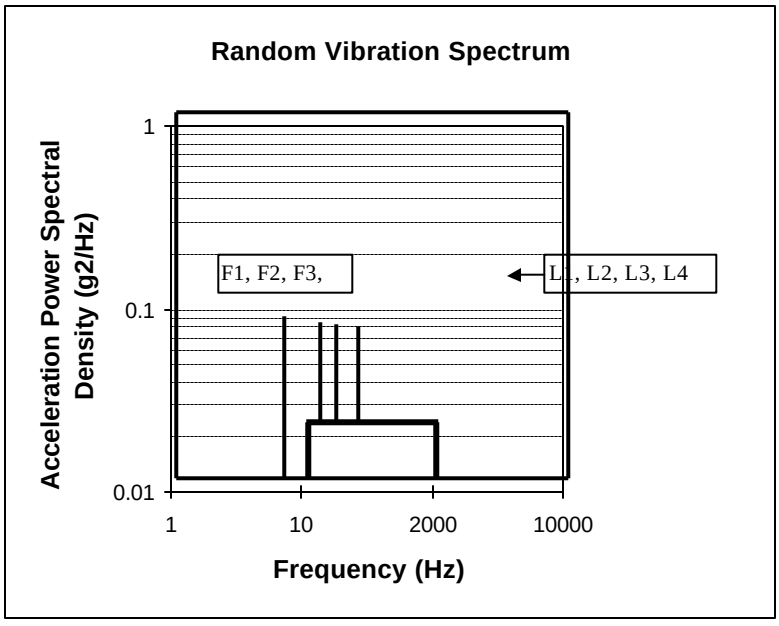


Figure 3. Random Vibration Spectrum

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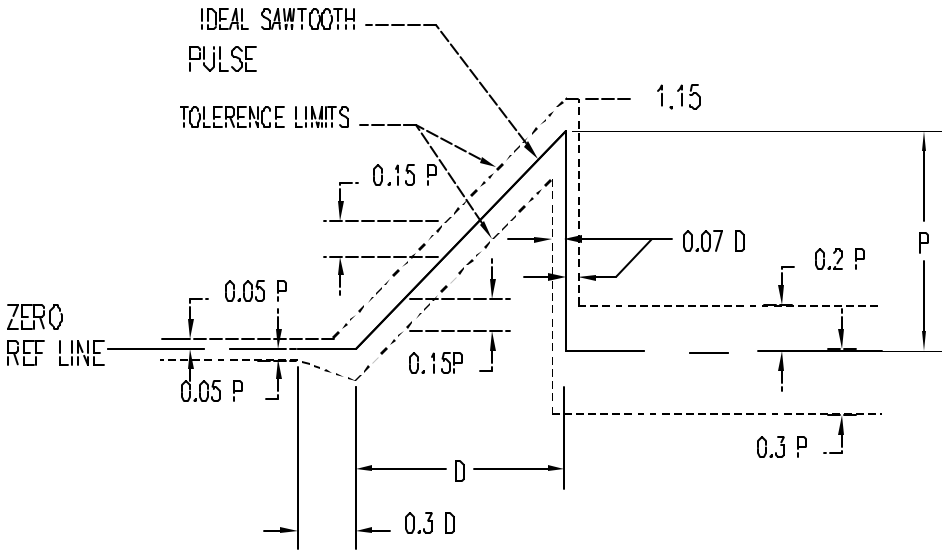


Figure 4. Terminal peak saw tooth shock pulse

4.3.5 Temperature-Shock/Humidity/Altitude (TS/H/A) cycling. The Thruster with protective caps installed shall be submitted to the TS/H/A cycling test outlined in MIL-D-23615, paragraph 4.8.4 with the following requirements:

a. The Thruster shall be supported in such a way that all areas are exposed to the prescribed atmospheric conditions at all times throughout the test. No Thruster shall be in contact with another Thruster during the cycling schedule shown in Table V.

b. Thirty thousand (30,000) feet shall be used in lieu of 70,000 feet for the altitude test.

c. This schedule shall be followed for a total of four (4) weeks (28 days) except that on the second and fourth weekend the soak time shall be from 1200 on Friday until 0800 on Monday at a temperature of 160°F at 95% RH. There shall be no interim withdraws of units.

d. Failure of any Thruster to comply with the requirements of 3.6.3 shall be cause for rejection of the lot and/or first article.

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Table V. Temperature shock/humidity/altitude cycling schedule

Monday	0800	Place Thrusters in chamber at 70°F at 50% relative humidity (RH).
	1200	Raise chamber temperature to 160°F and an RH of 95%. The chamber shall reach 160°F no later than 1300.
	1600	Remove the Thruster's from above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 30,000 feet (4.36 psi)
Tuesday	0800	Remove Thruster's from chamber and immediately place in chamber maintained at 70°F at 50% RH.
	1200	Remove the Thruster's from above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 30,000 feet (4.36 psi)
	1600	Remove Thruster's from above chamber and immediately place in a chamber maintained at 160°F at 95% RH.
Wednesday	0800	Reduce chamber temperature to 70°F and RH to 50%. The chamber temperature shall reach 70°F at 50% RH no later than 0900.
	1200	Raise chamber temperature to 160°F and an RH of 95%. The chamber shall reach 160°F no later than 1300.
	1600	Remove the Thruster's from above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 30,000 feet (4.36 psi)
Thursday	0800	Remove Thruster's from chamber and immediately place in chamber maintained at 70°F at 50% RH
	1200	Remove the Thruster's from above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 30,000 feet (4.36 psi)
	1600	Remove Thruster's from above chamber and immediately place in a chamber maintained at 160°F at 95% RH.
Friday	0800	Reduce chamber temperature to 70°F and RH to 50%. The chamber temperature shall reach 70°F at 50% RH no later than 0900.
	1200	Raise chamber temperature to 160°F and an RH of 95%. The chamber shall reach 160°F no later than 1300.
	1600	Remove the Thruster's from above chamber and immediately place in a chamber maintained at -65°F at standard ambient pressure.

4.3.6 Six-foot drop. The Thruster, with protective shipping caps removed, shall be submitted to the drop test outlined in paragraph 4.8.2 of MIL-D-23615 with the following requirements:

a. The Thruster shall be dropped from a height of 6 feet onto a 2 inch thick steel plate supported by reinforced concrete. One Thruster shall be used for each drop orientation. All impact orientations are given in figure 1.

b. After each Thruster is dropped, a visual inspection shall occur to document any external damage that occurred and to verify non-initiation of the Thruster.

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c. If it is determined that the damage sustained by any unit caused it to be unsafe for handling all testing shall be stopped.

d. Failure of any Thruster to comply with the requirements of 3.6.5 shall be cause for rejection of the lot and/or first article.

4.3.7 Functional performance. Sample Thruster for the functional performance test firings shall be conditioned for not less than 4 hours and no more than 24 hours at the specified temperature. The Thruster shall be removed from the conditioning chamber and fired in the appropriate test fixture (figure 5) within 5 minutes. If any Thruster is not fired within the 5 minute time period after removal, it shall be reconditioned at the specified temperature for an additional 2 hours and then tested. The Thruster shall meet the performance requirements of 3.6.5.1 or 3.6.5.2.

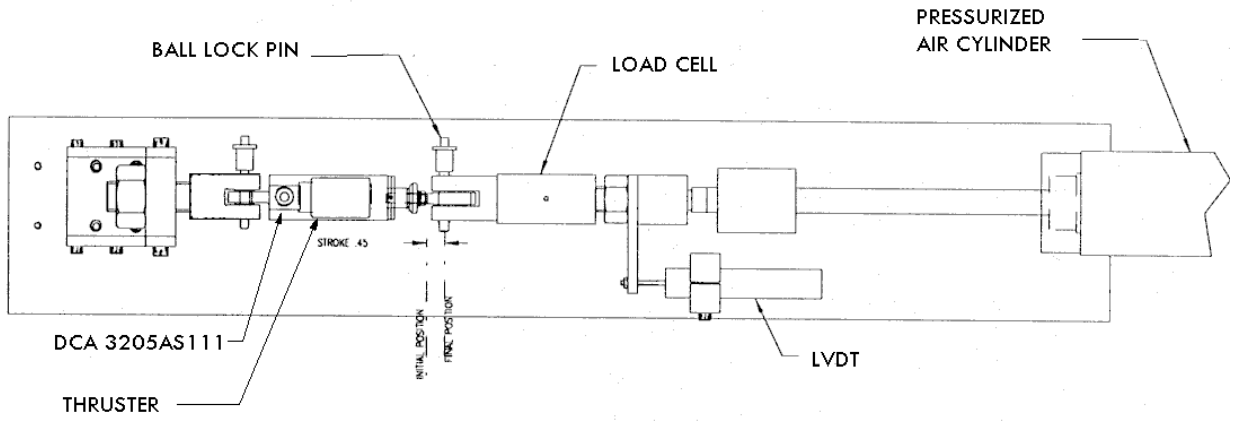


Figure. 5 Functional Test Firing Setup (17610 TN 172682-1)

4.3.7.1 Thruster actuation. The Thruster shall be functioned by a Detonating Cord Assembly (3205AS111) which incorporated a small gas generator in the out put tip. The Thruster shall meet the performance requirements of 3.6.5.1.

4.3.7.2 Peak thrust. The Thruster shall break the shear pin and overcome a 197-lbs. across the operational temperature range. The Thruster shall meet the performance requirements of 3.6.5.2.

4.3.7.3 Time to stroke. The time to stroke shall be measured from initiation of the Detonating Cord Assembly to full stroke at 0.45 inch.

4.3.8 Test Failure. If a test failure is attributable to an assigned cause, excluding the Thruster, the test shall be declared a no-test and repeated.

4.4 Acceptance criteria. The Thruster shall meet the requirements of 3.6 when tested as specified in 4.3.3 through 4.3.11. The acceptable number of defects is 0 and the rejection number is 1.

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5. PACKAGING

5.1. Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points packaging activity within the Military Department or Defense Activity, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department or Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that maybe helpful but is not mandatory.)

6.1 Intended Use. These Thrusters are components of the Emergency Egress Hatch Removal Systems installed in the MV-22, CV-22 and HV-22 variant aircraft. They are the mechanical thrusters using ballistic energy for a gas generator tip in 3205AS111 to rotate the seat drop mechanism. As the Emergency Egress Hatch Removal Systems is designed only for military V-22 Aircraft, there is no commercial application for the TCU-5A/A thruster.

6.2 Acquisition Requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced.

c. Whether a first article inspection is required and. if so, specify the test activity (see 3.1 and 4.2.1).

d. Markings if other than as specified (see 6.5.2.1).

e. If bar coding is not required on outer container (see 6.5.2.2).

f. Inspection conditions if other than specified in 4.2.

g. Production lot size (quantity) and test facility (see 4.2.3.1).

h. Configuration of the Thruster required (see 2.2.2).

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6.3 First Article. When a first article inspection is required, the contracting officer should provide specific guidance to bidders whether the item(s) should be a pre-production sample, a first article sample, a first production item, a sample selected from the first production items, standard production item from the contractor's current inventory (see 3.1) and the number of items to be tested as specified in 4.2.1. The contracting officer should also include specific instructions in acquisition documents of the first articles. Invitation(s) for bid(s) should provide that the U.S. Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the U. S. Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior U.S. Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Primary component. Primary components are all components in which a functional failure would result in a misfire or malfunction of the Thruster.

6.4.2 Flight critical part. A part, the single failure of which, during any operating condition, could cause the loss of aircraft or one of its major components, loss of control, or which may cause significant personnel injury, including during flight.

6.5 Packaging

6.5.1 Preservation, packaging and packing. Preservation, packaging, and packing should be in accordance with the requirements listed below. The contractor should also comply with all applicable transportation regulations.

6.5.1.1 Caps. Aluminum or plastic shipping caps should be installed in the port.

6.5.1.2 Inner package. One Thruster as prepared per 6.5.1.1 should be packaged in heat sealed vapor barrier bag or equivalent.

6.5.1.3 Outer pack. The Thruster packaged as described in 6.5.1.2 should be packaged in a container as described in the contract or purchase order.

6.5.2 Markings.

6.5.2.1 Normal marking. Unless otherwise specified in the contract or purchase order (see 6.2), the marking on the container should be as specified below. The specified markings should be applied to the containers in accordance with the applicable provisions of MIL-STD-129.

(1) NSN: 1377-01-499-2336

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(2) DODIC: JL02
(3) Thruster, Cartridge Actuated TCU-5A/A
(4) Drawing 3205AS117-2
(5) Quantity: 1
(6) Gross weight and cube
(7) Lot number in accordance with MIL-STD-1168
(8) Contract/purchase order number

6.5.2.2 Bar Coding. Unless otherwise specified in the contract or purchase order (see 6.2), the outer container should be bar coded in accordance with MIL-STD-129-1, Part 2. Additionally, the above-specified marking information should be applied to the containers in accordance with the provisions of MIL-STD-129.

6.5.2.3 Markings. The markings of the outer container as a minimum should include the NIIN and the lot number, shelf life expiration date, and quantity. The bar code label should be applied to the outer container.

6.6 Subject term (key word) listing.

TLX ®
Emergency Egress Hatch Removal Systems

Preparing activity: Navy-OS
(Project 1377-0216)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://www.dodssp.daps.mil.

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